Irene Langkilde-Geary

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South Jordan, UT 84095

Position: Research Scientist in NLP

Employment Experience

NLP Scientist/Engineer, ClearWhite Model Programming (self-employed), March 2023-present
Applied a novel smoothing technique to p-o-s tagging and dependency parsing using logistic regression and clustering. An eventual goal was to support increased automation of the language data annotation process and improved accuracy on a variety of applications.

Senior Data Scientist, **Aktify**, June 2021-March2022
Developed novel probabilistic word embeddings trained on 2.4T of parsed web-crawled data with the intention of applying them to multi-intent detection of utterances in conversational AI (written in Julia). Contributed to design of dialog model and dialog management architecture.

Natural Language Processing Scientist/Engineer, **Lifepod**, January 2019-June 2021.
Architected and implemented custom dialog manager for proactive routines, with patentable innovations. Developed voice-based conversational agents using Google's Dialogflow and Amazon's Alexa skills. Wrote custom intent-detection logic on top of syntactic analysis APIs (which provided dependency-style syntactic analyses and machine-learned sentiment scores) to classify utterance sentiment much more accurately than ML alone. Wrote batch testing harness (Golang).

Natural Language Processing Scientist/Engineer, **PipelineEquity**, February-April 2020.
Trained text categorization models using AWS Comprehend/Sagemaker and SpaCy.

Research Scientist/Software Developer, **Cobalt Speech and Language**, July 2015-Sept 2018.
Trained custom domain-specific language models for use with open-source Kaldi speech recognizer. Developed production ASR engine extension of Kaldi (C++ code). Wrote and adapted Perl, Python, and Bash scripts to clean and prepare large data sets. Generated specialty sub-domain language models by writing context-free grammar rules and compiling them with Thrax. Acted as a liaison with clients.

Research Hobbyist/At-Home Mother of Five, **Independent**, November 2008-2015. Developed omni-directional natural language parser and generator, integrating heterogeneous reasoning techniques including logical constraints and probabilistic inference (initially in Mozart/Oz constraint programming language, then JuliaLang). Developed novel smoothing technique and more effective variant of logistic regression for text classification. Developed framework for interactive ML.

Research Fellow, **University of Brighton,** UK, June 2007-November 2008.
Developed a template realizer for the REG (Referring Expression Generation) Challenge. Compared several manual and semi-automatic techniques for generating text from data, such as weather forecasts

and wikipedia fact tables.

Consultant, **Southwest Research Institute**, Ogden, UT, April 2006-May 2007.
Built a prototype system to automate the generation of some Learning Object Metadata (LOM) Schema for online courses.

Assistant Professor**, Brigham Young University**, UT, December 2002-August 2006.
Theses Supervised
2004, Masters, Dan Su, “A Target-Dominant Approach to Machine Translation.”
2006, Masters, Thomas Packer, “Surface Realization Using a Featurized Syntactic Statistical Language Model.”
2007, Masters, Robert Van Dam, “Adapting ADtrees For Improved Performance on Large Datasets with High Arity Features.”

Courses Taught
CS 330 Programming Languages CS
470 Artificial Intelligence CS 601R
Natural Language Processing

Education

PhD thesis (2002), “Automatic Sentence Generation Using a Hybrid Statistical Model of Lexical Collocations and Syntactic Relations.”

Ph.D. Computer Science, **University of Southern California**, 2002
M.S. Computer Science, **University of Southern California**, 1999
B.S. Computer Science, **Brigham Young University**, 1995, *cum laude*

Awards and Honors

- Test of Time Award (1998), Association of Computational Linguistics, received July 2023
- Dean's Merit Fellowship for Doctoral Studies at USC, 1995
- National Merit Scholarship at BYU, 1989

Patents

- US 6,751,591: Method And System For Predicting Understanding Errors In A Task Classification System, 06-15-2004.

Skills

Natural Language Processing: conversational AI, sentiment analysis, information extraction, machine translation, p-o-s tagging, parsing/generation, programmatic metadata annotation, semi-supervised interactive probabilistic machine learning and model building.

Conversational AI: custom designed a system for Lifepod; also experienced with Google Dialogflow, *Amazon Alexa (and Lex); Rasa.*

NLP tools: Stanford_CoreNLP, NLTK, GATE, SpaCy.

Machine Learning tools: AWS Comprehend, AWS Sagemaker, Tensorflow.

Machine Learning methods: Logistic/probabilistic Regression, DecisionTree, K-means clustering, NaiveBayes.

Programming Languages/tools: Julia, Golang, C/C++, Perl, Python, Lisp, Javascript, HTML, CSS, Make, Gdb, Valgrind, Zsh, Bash, MySQL, Scons, Git, VSCode, Atom, Emacs.

DevOps: Agile Jira, AWS: EC2, S3, Athena, Cloudwatch, Jenkins; Datadog.

Publications Available Upon Request

References Available Upon Request